

## Terraform Task-2

Task Title: Create 2 EC2 instances on 2 different regions (Mumbai & Hyderabad) and install Nginx using Terraform.

1. Local Environment & Terraform Verification The local environment path variable was updated to map the location of the binary. The setup was confirmed using the global version check invocation: PS C:\Users\HP> terraform -v Output: Terraform v1.15.8 on windows\_amd64 Status: SUCCESSFUL - CLI fully functional globally.

```
PS C:\Users\HP> $env:Path += ";C:\terraform"
PS C:\Users\HP> $env:Path += ";C:\terraform"
PS C:\Users\HP> terraform -v
Terraform v1.15.8
on windows_amd64
PS C:\Users\HP>
```

```
PS C:\terraform-aws-project> terraform init
Initializing the backend...

Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.59.0...
- Installed hashicorp/aws v6.59.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

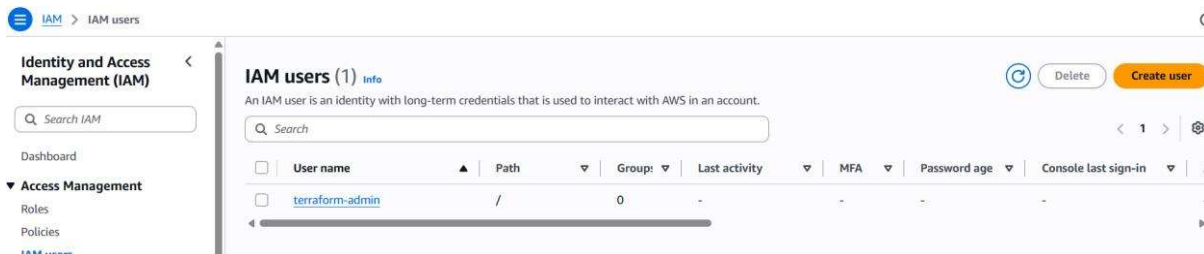
If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\terraform-aws-project> |
```

2. Project Directory Workspace Initialization A dedicated configuration workspace was created inside C:\terraform-aws-project. The provider workspace initialization sequence was executed to retrieve structural plugins: PS C:\terraform-aws-project> terraform init Logs Captured: Finding latest version of hashicorp/aws... Installing hashicorp/aws v6.59.0... Terraform has been successfully initialized!

### 3. Authentication Safeguards & Authorization Fixes

Initial planning failed due to missing cloud access configuration (Error: No valid credential sources found). Root user keys were flagged by AWS IAM console security checks as non-compliant with security paradigms.

Remediation Action: A dedicated non-root IAM user (terraform-admin) was created with Administrator Access policies attached. Credentials were safe-linked via terminal: PS C:\terraform-aws-project> aws configure AWS Access Key ID: AKIAXXXXXXXXXXXXXXXXXX Default region name: ap-south-1 Default output format: json



```
PS C:\Users\HP> cd C:\terraform-aws-project
PS C:\terraform-aws-project> aws configure

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session
in'.

AWS Access Key ID [None]: AKIAXGOF4RZDYPS6D5PD
AWS Secret Access Key [None]: js0gVc0uCZFbWYfnHotwSsW7T4MTd2L573rvRa2m
Default region name [None]: ap-south-1
Default output format [None]: json
```

### 4. Infrastructure Execution Dry-Run

With credentials successfully bound, the execution blueprint mapping was compiled safely against target providers:

```
PS C:\terraform-aws-project> terraform plan
data.aws_ami.ubuntu_hyderabad: Reading...
data.aws_ami.ubuntu_mumbai: Reading...
data.aws_ami.ubuntu_hyderabad: Read complete after 1s [id=ami-00cae351afad5d61e]
data.aws_ami.ubuntu_mumbai: Read complete after 2s [id=ami-0aa761682283b4cc8]

Terraform used the selected providers to generate the following execution plan. Resource actions are indi
following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.web_hyderabad will be created
+ resource "aws_instance" "web_hyderabad" {
  + ami              = "ami-00cae351afad5d61e"
  + arn              = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone = (known after apply)
  + disable_api_stop   = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized     = (known after apply)
  + enable_primary_ipv6 = (known after apply)
```

```

PS C:\terraform-aws-project> terraform apply
data.aws_ami.ubuntu_hyderabad: Reading...
data.aws_ami.ubuntu_mumbai: Reading...
data.aws_ami.ubuntu_hyderabad: Read complete after 1s [id=ami-00cae351afad5d61e]
data.aws_ami.ubuntu_mumbai: Read complete after 1s [id=ami-0aa761682283b4cc8]

Terraform used the selected providers to generate the following execution plan. Resources to be
added, changed, or destroyed are highlighted in blue.

following symbols:

```

## 5. Infrastructure Assembly Execution

The live provision signal was sent. Networks, security parameters, and compute resources generated cleanly inside 14 seconds across both isolated regions.

```

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_security_group.sg_mumbai: Creating...
aws_security_group.sg_hyderabad: Creating...
aws_security_group.sg_hyderabad: Creation complete after 4s [id=sg-09b1f2a56a65ff540]
aws_security_group.sg_mumbai: Creation complete after 4s [id=sg-03a7fdd3c1c92fe26]
aws_instance.web_hyderabad: Creating...
aws_instance.web_mumbai: Creating...
aws_instance.web_hyderabad: Still creating... [00m10s elapsed]
aws_instance.web_mumbai: Still creating... [00m10s elapsed]
aws_instance.web_hyderabad: Creation complete after 14s [id=i-0cad8ca84f056f342]
aws_instance.web_mumbai: Creation complete after 14s [id=i-0c2b40e3db15c8d76]

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.

Outputs:

```

## 6. Dynamic URL Verification Metrics

Live public server IPs generated by the provider outputs were mapped directly into the system web explorer layout:

| Target Region          | Public Endpoint IP | Render Verification Response Status                                                |
|------------------------|--------------------|------------------------------------------------------------------------------------|
| Mumbai (ap-south-1)    | 3.110.189.49       | Renders Default 'Welcome to nginx!' page successfully.                             |
| Hyderabad (ap-south-2) | 18.60.60.148       | Renders Targeted Custom 'Welcome to Nginx in Hyderabad Region!' text successfully. |



## Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to [nginx.org](http://nginx.org).  
Commercial support is available at [nginx.com](http://nginx.com).

*Thank you for using nginx.*